

TEAM CONTRIBUTION REPORT

JAMUNA VP (DATA ENGINEER)

Overall Role Perspective

I functioned as the **Data Engineer** of the team and played a continuous, high-impact role throughout the lifecycle of the project. My responsibilities extended beyond routine data handling and included planning support, data architecture preparation, preprocessing strategy, feature creation, backend support for dashboard development, quality assurance, and report completion. While every member contributed in their own area, my involvement was deeply connected to almost every technical phase of the project, making my contribution foundational to the success of the final outcome.

WEEKLY CONTRIBUTION ANALYSIS – JAMUNA VP

Week 1 – Project Initiation, Planning, and Problem Understanding

During the initial phase of the project, I actively contributed to establishing the direction of the study. I participated in discussions regarding the relevance of cryptocurrency as a modern financial domain and helped shape the project around the problem of volatility and unpredictable price movement. My contribution in this stage involved understanding what type of data would be required, what variables would be meaningful, and how technical analysis could later be supported through proper data preparation.

I also helped break down the project into practical modules such as:

- Dataset collection
- Cleaning and preprocessing
- Exploratory analysis
- Dashboard creation
- Storytelling and reporting

By helping organize the project structure early, I contributed to a smoother workflow in later weeks.

Week 2 – Data Acquisition and Raw Dataset Structuring

In the second week, I played a major role in sourcing and organizing the required datasets. The project depended on two different forms of data: structured financial data and semi-structured social media data. I contributed significantly in obtaining and arranging these sources into formats suitable for analysis.

Key Responsibilities:

- Assisted in collecting Bitcoin historical price data from trusted sources such as CoinMarketCap.
- Helped gather cryptocurrency-related tweet data from Kaggle sources.
- Organized multiple files into structured datasets.
- Checked row counts, column availability, and relevance of attributes.
- Ensured proper file storage and naming conventions for team use.

Technical Value Added:

- This stage was critical because poor raw data management can affect all downstream tasks.
- My contribution ensured that the team began with clean source files and a reliable dataset foundation.

Week 3 – Data Cleaning, Error Correction, and Preprocessing

This was one of the most important phases of the project, and my role was especially valuable here. The report identifies multiple data issues such as missing values, formatting errors, noise, and unstructured records. I contributed extensively to resolving these challenges and converting raw data into a usable analytical dataset.

Major Contributions:

- Detected null and missing values in numeric and text columns.
- Applied mean imputation techniques for numerical features.
- Removed incomplete or unusable tweet rows.
- Corrected delimiter issues where records were merged into single columns.
- Removed unwanted symbols such as \$, %, and commas from financial fields.

- Converted string-based columns into numeric datatypes.
- Standardized date/time columns for time-series analysis.
- Helped reduce noise in text-based sentiment data.

Impact:

- Without this preprocessing stage, later analysis would have been inaccurate or impossible.
- My work directly improved data quality, consistency, and trustworthiness.

Week 4 – Feature Engineering and Data Transformation

After cleaning the data, I contributed to transforming the dataset into a more insightful and model-ready form. Instead of relying only on raw columns, I helped generate meaningful derived variables that improved analysis quality.

Features Supported:

- Price Change = Close – Open
- Price Movement = Up / Down classification
- High-Low Range for volatility measurement
- Open-Close Difference
- Volume to Market Cap Ratio

Transformation Work:

- Applied normalization techniques.
- Supported scaling methods such as standardization.
- Helped address skewness in variables like trading volume.
- Prepared data ranges suitable for comparison across variables.

Why It Mattered:

- These engineered features made the project more analytical and professional.
- They enabled deeper insights beyond simple raw price reporting.

Week 5 – Exploratory Data Analysis and Pattern Discovery

In this stage, I helped extract insights from the processed data. My role focused on connecting clean data to meaningful observations that could support the project's objectives.

Areas of Contribution:

- Supported analysis of closing price distributions.
- Helped study trading volume spikes and skewed patterns.
- Assisted in correlation analysis between open, high, low, close, and volume.
- Contributed to time-series trend understanding.
- Helped interpret clustering ranges in market behavior.
- Participated in linking social media activity trends with market movements.

Analytical Thinking:

- Rather than only processing data, I also contributed to interpreting what the numbers meant in practical investment or volatility terms.
- This strengthened the quality of the final insights chapter.

Week 6 – Dashboard Backend Preparation and Visualization Support

Although the visualization layer may have been team-driven, my contribution was highly important behind the scenes. Dashboards depend heavily on structured data pipelines, and I helped ensure that the backend data feeding the visuals was correct and efficient.

Key Contributions:

- Prepared final cleaned datasets for Power BI import.
- Helped define table relationships and field mappings.
- Verified measures used in KPI cards.
- Supported metric calculations such as average price, max price, min price, and total volume.

- Ensured consistency between raw values and displayed dashboard results.
- Assisted in testing filter behavior and chart responsiveness.

Significance:

- Even excellent visuals lose value if the underlying data is wrong.
- My work ensured that the dashboard was dependable and presentation-ready.

Week 7 – Validation, Quality Assurance, and Final Optimization

Before project submission, I continued contributing through systematic verification and refinement. This stage focused on ensuring the final output was technically accurate and professionally polished.

Contributions:

- Cross-checked processed outputs with original source datasets.
- Verified feature columns and derived calculations.
- Helped detect inconsistencies in charts or summary metrics.
- Supported correction of formatting and labeling issues.
- Improved data presentation quality.
- Assisted in final testing of end-to-end workflow.

Outcome:

- My quality assurance efforts reduced the risk of errors during evaluation and strengthened the credibility of the project.

Week 8 – Documentation, Report Writing, and Final Submission

In the concluding phase, I contributed to the academic presentation of the project. Since technical work must also be communicated clearly, my involvement in report preparation added strong value.

Areas Supported:

- Contributed content for preprocessing and methodology chapters.
- Helped organize sections logically.
- Assisted in refining explanations for feature engineering and analysis.
- Participated in proofreading and consistency checks.
- Coordinated during final report assembly and submission.

Final Value:

- My role ensured that the technical depth of the project was properly reflected in the final report and not lost in documentation gaps.

COMPREHENSIVE FINAL SUMMARY – JAMUNA VP

I was one of the most technically significant contributors to the Cryptocurrency Volatility Tracker project. As Data Engineer, I was involved from the earliest planning stage to final submission. My work spanned data collection, cleaning, transformation, feature engineering, validation, backend dashboard preparation, and technical documentation. I consistently enabled the team to move from raw, inconsistent datasets to a polished analytical product.

My greatest strength was not only executing assigned tasks, but ensuring that every later stage of the project had a strong and reliable data foundation. Because modern analytics projects depend first on data quality, my contribution had a multiplier effect across the entire report. The successful completion of the project was strongly supported by my continuous technical involvement, structured approach, and dependable execution across all weeks.